

HAILUOTO KESKIKYLA, Finland

WMO: 028740

Lat: 65.020N

Lon: 24.727E

Elev: 20

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-11.0	-21.7	1.7	-16.6	-15.5	2.4	-11.0	21.0	31.3	19.0	31.3	1.7	110	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	77.1	65.1	73.3	63.3	70.1	61.7	67.4	74.3	65.2	71.0	63.1	68.6	6.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	92.3	70.5	62.8	85.7	67.9	60.7	79.4	66.3	31.8	74.9	30.2	71.1	28.6	68.6	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	16.2	14.3	-19.4	80.3	7.5	5.3	-24.8	84.1	-29.2	87.1	-33.4	90.1	-38.8	93.9		
			-19.7	69.7	7.4	3.2	-25.1	72.1	-29.4	74.0	-33.6	75.8	-39.0	78.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	16.4	16.7	23.6	33.8	45.6	53.6	60.4	57.3	49.2	38.0	30.7	24.6
	DBStd	17.28	13.15	13.41	9.60	5.93	7.44	6.06	5.66	5.75	5.84	7.00	8.27	11.43
	HDD50	5320	1041	932	818	487	179	28	1	10	84	375	578	787
	HDD65	10029	1506	1352	1283	937	602	345	165	247	474	838	1028	1252
	CDD50	803	0	0	0	0	43	138	324	236	60	2	0	0
	CDD65	36	0	0	0	0	1	4	23	8	0	0	0	0
	CDH74	283	0	0	0	0	12	46	179	46	0	0	0	0
	CDH80	45	0	0	0	0	0	9	29	7	0	0	0	0
Wind	WSAvg	6.6	6.2	6.6	7.1	6.6	7.2	6.7	5.5	5.8	6.3	6.8	6.9	7.0
Precipitation	PrecAvg	20.5	1.5	1.3	1.2	1.1	1.5	1.7	2.5	2.2	1.9	2.0	1.9	1.7
	PrecMax	27.3	2.9	2.7	3.2	3.1	3.3	4.0	4.7	4.6	3.5	3.8	3.3	3.4
	PrecMin	14.6	0.6	0.6	0.2	0.3	0.2	0.1	0.1	0.2	0.4	0.3	0.4	0.5
	PrecStd	3.3	0.7	0.7	0.7	0.6	0.7	1.1	1.2	1.0	0.9	0.8	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	39.2	42.0	57.1	75.4	80.5	83.7	79.1	67.2	55.0	47.8	41.2
		MCWB	35.9	36.6	37.3	45.8	59.0	63.8	67.6	66.0	59.0	51.4	45.7	39.5
	2%	DB	35.6	37.0	38.8	51.0	69.4	72.5	78.6	73.6	62.8	51.9	45.6	39.6
		MCWB	34.6	35.1	35.2	42.3	57.2	60.8	66.5	64.0	57.2	49.3	44.1	38.1
	5%	DB	34.0	34.8	37.0	46.2	63.1	68.4	74.9	70.3	60.2	49.7	43.2	38.1
		MCWB	33.2	33.4	34.4	40.0	52.3	58.0	64.6	62.4	56.0	47.6	41.9	36.7
	10%	DB	32.6	32.9	35.2	42.7	58.4	64.9	71.8	67.5	58.1	48.0	40.9	36.6
		MCWB	32.0	32.0	33.1	37.8	49.8	56.4	63.3	61.4	54.8	46.1	39.7	35.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	37.2	38.1	47.7	61.0	67.0	71.0	69.1	60.9	52.5	46.5	40.3
		MCDB	37.1	38.8	41.1	53.6	72.4	77.4	77.8	76.0	64.7	54.3	47.2	40.7
	2%	WB	34.9	35.3	36.0	43.0	57.3	62.9	68.6	66.1	58.4	49.9	44.1	38.4
		MCDB	35.4	36.7	37.9	49.1	67.5	69.7	75.7	70.9	61.5	51.4	45.3	39.2
	5%	WB	33.3	33.7	34.7	40.4	53.8	60.2	66.5	64.1	56.7	48.1	42.0	36.9
		MCDB	33.8	34.6	36.4	45.9	61.7	65.9	72.7	68.3	59.2	49.5	42.9	38.0
	10%	WB	32.1	32.2	33.5	38.3	50.8	57.8	64.4	62.1	55.3	46.2	39.9	35.5
		MCDB	32.6	33.0	35.0	42.1	56.6	63.5	70.5	66.2	57.7	47.8	40.8	36.4
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.4	12.9	13.4	15.4	15.0	15.5	14.8	12.0	9.1	7.2	9.1
		MCDBR	8.1	8.3	10.4	20.6	24.9	21.7	21.5	20.2	15.2	9.9	7.3	7.0
		MCWBR	8.1	7.4	8.0	13.4	13.2	11.6	10.9	11.2	10.5	8.2	7.0	6.3
	5% WB	MCDBR	8.2	7.9	8.8	18.4	23.1	17.7	18.6	16.4	13.3	9.2	7.3	6.8
		MCWBR	8.2	7.3	7.1	12.4	13.5	10.5	10.4	10.6	9.8	8.0	7.0	6.2
Clear-Sky Solar Irradiance	taub	0.209	0.276	0.305	0.343	0.347	0.350	0.370	0.354	0.331	0.308	0.344	0.354	
	taud	1.951	2.097	2.141	2.190	2.410	2.483	2.450	2.486	2.526	2.353	2.545	2.590	
	Ebn at Noon	93	189	240	260	273	275	264	258	239	183	85	36	
	Edh at Noon	7	19	29	35	31	30	30	26	21	15	7	4	
All-Sky Solar Radiation	RadAvg	34	196	663	1202	1575	1764	1629	1205	660	239	49	10	
	RadStd	3	38	67	96	149	140	146	113	67	39	8	1	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page